Completed

Non-Preemptive Shortest Job First CPU Scheduling Algorithm

Write a C program to implement the Non-Preemptive Shortest Job First (SJF) CPU Scheduling Algorithm. The program should accept the number of processes, their Arrival Times (AT), and Burst Times (BT). At any given time, the process with the minimum burst time among the arrived processes must be executed first. Calculate the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) for each process and find the overall average TAT and WT.

Core Logic

Selection: Out of all processes that have arrived by the current time, the one with the minimum Burst Time is chosen.

Formulae:

- $Turnaround Time (TAT) = Completion Time (CT) - Arrival Time (AT)$
- $Waiting Time (WT) = Turnaround Time (TAT) - Burst Time (BT)$

Input Format

1. The first line of input contains an integer representing the number of processes

Select Language : C (GCC 9.2.0)

```c
#include <stdio.h>
int main() {
    int n, i, completed = 0, current_time = 0;
    printf("Enter number of processes: \n");
    scanf("%d", &n);
    int AT[n], BT[n], CT[n], TAT[n], WT[n], visited[n];
    for(i = 0; i < n; i++) {
        printf("Enter Arrival Time and Burst Time for P%d: \n", i + 1);
        scanf("%d %d", &AT[i], &BT[i]);
        visited[i] = 0;  // Mark all processes as not completed
    }
    float totalTAT = 0, totalWT = 0;
    while(completed < n) {
        int minBT = 9999;
        int index = -1;
        for(i = 0; i < n; i++) {
            if(AT[i] <= current_time && visited[i] == 0) {
                if(BT[i] < minBT) {
                    minBT = BT[i];
                    index = i;
                }
            }
        }
        if(index == -1) {
```

Test Cases 2

Run Sample Cases

Run All Cases

Completed

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```c
            if(AT[i] <= current_time && visited[i] == 0) {
                if(BT[i] < minBT) {
                    minBT = BT[i];
                    index = i;
                }
            }
        }
        if(index == -1) {
            current_time++;
        } else {
            CT[index] = current_time + BT[index];
            current_time = CT[index];
            TAT[index] = CT[index] - AT[index];
            WT[index] = TAT[index] - BT[index];
            totalTAT += TAT[index];
            totalWT += WT[index];
            visited[index] = 1;
            completed++;
        }
    }
    printf("\nAverage Turnaround Time = %.2f\n", totalTAT / n);
    printf("Average Waiting Time = %.2f\n", totalWT / n);
    return 0;
}
```

Test Cases 2

Run Sample Cases

Run All Cases